Ars Logica

Doctrine Primers for the Exercises

Logic as an Art, Exercises I through XXI

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EXERCISE I

The Predicables

A predicable is a way that a general word can be said of a subject. Porphyry set out five. The same word can be a different predicable of a different subject, so ask about this predicate of this subject.

The five predicables

Genus

The wider kind. It is said of many things that differ in species, and it answers "What is it?" with the wider nature. "Animal" is the genus of man and of the horse.

Species

The narrower kind. It is said of individuals, and it answers "What is it?" completely. Socrates is a man; Fido is a dog.

Difference

The mark that divides the genus and makes the species. It answers "What sort of thing is it, in its very nature?" "Rational" divides animal and makes man. Genus plus difference gives the species.

Property

Not part of what the thing is, but it follows from what the thing is. It belongs to every member of the species, to that species alone, and always. The power to laugh is the property of man.

Accident

What a subject may have or lack while it stays the same thing. Socrates may be pale or tanned. A separable accident comes and goes (sitting). An inseparable accident never leaves (the blackness of a raven), but it is still no part of what the thing is.

Two questions sort them

	Said of more than the subject	Said of the subject alone
	Genus	Difference

Part of what it is		
Not part of what it is	Accident	Property

The triangle: "figure" is its genus; "three-sided" is its difference; "having its angles equal to two right angles" is its property; "drawn in chalk" is an accident.

The swap test

A property swaps with its species: every man is able to laugh, and whatever is able to laugh is a man. A genus does not swap: every man is an animal, but not every animal is a man. An accident does not swap either: every raven is black, but much that is black is not a raven.

Power or act?

A small change of words changes the answer. "Man is able to laugh" gives a property, because the power is always there. "This man is laughing" gives an accident, because the laughing comes and goes. So too: able to learn grammar (property); knows grammar (accident of this boy).

Genus and species are relative

In the tree of Porphyry, a kind in the middle is a genus to what is below it and a species to what is above it. "Animal" is a genus of man and a species of living body. A far-off genus is still a genus: "An oak is a living thing" gives a genus. Substance is the highest genus and never a species. Man is the lowest species and never a genus. Socrates is neither: things are said of him.

Why five? Some predicates tell what a thing is (genus, species, difference); one always goes with it without being what it is (property); some just happen to be true of it (accident). So there are five. (In "Man is a species," "man" stands for the nature as known, not any man.)

EXERCISE II

The Categories

Aristotle set out ten categories. They are the highest kinds under which anything said of a thing falls. The exercise asks under which category a predicate falls.

Substance and the nine accidents

Substance exists in its own right, not in another: a man, a horse, a stone. The other nine are **accidents**, which exist only in a substance or affect it. So all the others depend on substance.

Primary substance is the individual, such as this man. **Secondary substance** is its species or genus, such as man or animal; it is said of the individual ("Socrates is a man"). The mark most proper to substance: one and the same substance can take on contraries by its own change, pale at one time and dark at another. Substance has no contrary and no more and less.

The ten, with examples

Category	What it is	Examples
Substance	exists in its own right	a man, a horse
Quantity	from matter; "how much?"	two feet long, the number ten
Quality	from form; "of what sort?"	white, hot, just
Relation	a bearing toward another	double, half, a master
Action	what it does to another	cutting, burning
Passion	what is done to it	being cut, being burnt
When	measures it in time	yesterday, now
Where	the place it is in	in the marketplace
Posture	how its parts are arranged in the place	he sits, he lies
Having	what it has on it	he is shod, he is armed

Quantity is either discrete (a number) or continuous (a line, a surface).

Why there are ten (St. Thomas)

Each accident is related to substance in a different way.

- Some are **in** the substance. If absolutely, they come from its matter (**quantity**) or from its form (**quality**). If as a bearing toward something else, they are **relation**.
- Some **affect** the substance by being in it: what it does is **action**; what is done to it is **passion**.
- Some affect it **from outside**. If they measure it in time, that is **when**. If they measure it in position, that is **where** (the place) or **posture** (the arrangement of its parts in that place). If they do not measure it but are something it has on it, that is **having**.

Easy mistakes

- Knowledge and virtue, as lasting states of the soul, are **quality** (a "habit" in the sense of a lasting quality), not the category of having.
- **Where** is the place itself; **posture** is the arrangement of the parts in it. "He sits" is posture, not an action going on.
- **Relation** cannot be understood without its other: no double without half, no master without a servant.
- The categories sort what things are; the predicables are relations things have because they are known.

Division

Division is "a statement that distributes a thing into its members or a term into its meanings" (John of St. Thomas). Division leads to definition: we divide a genus by differences until a single species is reached.

The four kinds of division

Why there are kinds: there are different kinds of whole, so there are different kinds of division.

Logical division

Divides a genus into its species, marked off by opposed differences. Animals into the rational and the irrational; musical instruments into strings, winds, and percussion.

Integral division

Divides a whole into the parts that make it up, and no part is called by the name of the whole. A book into its chapters (no chapter is a book); a house into foundation, walls, and roof.

Division by powers

Divides a soul, or one of its powers, by what it is able to do. The soul into its power to know and its power to desire; prudence into taking counsel, judging, and commanding.

Accidental division

Sorts a subject by features it can gain or lose without becoming a different thing. Apples into the red and the green.

Do not confuse: logical division is not **essential division**, which divides an essence into its parts (man into body and soul, or into animal and rational). And a division by powers is not integral: a paw is a bodily part, not a power of the soul.

The four rules

Whatever its kind, a sound division keeps four rules.

1. **Cover the whole.** Leave nothing out. Animals into mammals and birds is **incomplete**, since the fish are left out.

2. **No overlap.** Nothing may fall under two members at once.
3. **Every member belongs to the whole.** A tree into root, trunk, and the soil fails, since the soil is no part of the tree.
4. **One basis.** Shoes into leather, canvas, and left-footed sorts by material and by foot at once, so the members cut across each other.

Dividing "living thing" into plant and animal is sound: it is complete, does not overlap, and is sorted on one basis.

Four questions to ask

Does anything in the whole fall under no member? Does anything fall under two? Is any member outside the whole? Are the members sorted on one basis? A division can break more than one rule at once, and then each true answer is correct.

Why the rules are so

To divide is to sort out the whole so that everything in it has exactly one place. If something is left out, the whole is not divided. If something has two places, the members are not really different. If a member is outside the whole, it is not dividing that whole. If two ways of sorting are mixed, the places cross each other.

EXERCISE IV

Definition

A definition is "a sentence that sets forth the nature of a thing or the meaning of a term" (John of St. Thomas). The tradition counts four kinds and gives five rules.

The four kinds of definition

Nominal

Gives what the name means. "Philosopher" means a lover of wisdom; "hippopotamus" means river-horse. It comes first, because we cannot ask what a thing is until we know what its name refers to.

Essential

Says what the thing is, by its nearest genus and the difference that marks it off from everything else in that genus. Man is a rational animal.

Descriptive

Points the thing out by a property or a telltale feature, without saying what it is. Man is the animal that can laugh; the bee is the insect that makes honey.

Causal

Points the thing out by its outside causes: what made it, or what it is for. A clock is an instrument made to measure time; a scar is the mark left by a healed wound.

A definition by matter and form is **not** causal, because matter and form are inside the thing. It is a kind of essential definition, called the **physical** definition.

Why these four: we may explain the word (nominal), say what the thing is (essential), point it out by a mark that goes with it (descriptive), or point it out by where it came from or what it is for (causal).

The rules of definition

1. **Fit exactly.** Not too broad, not too narrow. The definition must be true of exactly the things the word is true of ("convertible"). "A triangle is a figure" is too broad. "A triangle is a figure with three equal sides" is too narrow.
2. **Be clearer than the thing defined.** So no metaphor. "A triangle is geometry's arrowhead" and "a window is the eye of the house" please but explain nothing.
3. **No circle.** Do not use the word being defined. "A triangle is a triangular figure" teaches nothing, because nothing is made clearer by itself.
4. **Say what it is, not what it is not,** wherever a positive account is available. "A line is length without breadth" is allowed, because no positive account is at hand.
5. **Be brief.** The nearest genus and the difference, and nothing more. "A triangle is a three-sided figure drawn with a ruler" and "Man is a rational animal able to laugh" carry wasted words.

"A triangle is a three-sided figure" keeps every rule.

Why the rules are so

A definition is meant to make a thing clear and to mark it off from everything else. If it covers too much or too little, it does not mark the thing off. If it is dark, circular, or only negative, it does not make the thing clear. Extra words add nothing and spoil it.

Diagramming Propositions

A Venn diagram is a picture of what a proposition says. Venn diagrams are a modern aid (John Venn, 1880). They help us see, but they prove nothing by themselves; the reasoning is done in words.

The four standard forms

Letter	Form	Diagram
A	All S are P	shade the part of S outside P
E	No S are P	shade the overlap
I	Some S are P	× in the overlap
O	Some S are not P	× in the part of S outside P

The marks

- **Shading** means the region is empty.
- **×**: the region is marked (for I, something is here). O itself carries no existential import, so for O the × only marks the region.
- A **plain** region is left open; the diagram says only what is marked.
- A **⊗** means something is there because an affirmative universal asserts that its subject exists.

The universals (A and E) shade a region empty, and the particulars (I and O) place an ×: for I it says that something is there, and for O it only marks the region.

Existence ("existential import")

In the older logic, which this course follows, an affirmative is true only if its subject stands for something that exists. So "All unicorns are white" is false, and "All S are P" yields "Some S are P." A negative carries no such existence, so "No unicorns are white" is true, and O carries none either: "Some unicorn is not white" is true. The ⊗ may be added to A in the overlap, but it is not required unless

an argument needs it. E and O never take a ⊗. I shows its existence with an ×, not a ⊗; for O the × only marks the region.

Finding the form in ordinary words

- A: "Every S is P," "Whatever is S is P."
- E: "S are never P," "There is no S that is P."
- I: "There are S that are P," "A few S are P."
- O: "Not all S are P," "Not every S is P."
- "Only P are S" means "All S are P."

Keep the subject's circle and the predicate's circle straight, because "All S are P" does not say "All P are S."

Two premises on three circles

The third circle cuts each region in two, so each universal premise shades two cells. Shade the universal premises first; then put each × in the one cell of its region that the shading leaves open. Draw only what the two premises say, with no ⊗.

Why

A universal says that nothing is in some place, so we empty it. I says that something is there, so we mark it; for O the × only marks the region, since O itself carries no existential import. Shading first shows where the × is forced to go.

Opposition, Equipollence, and Conversion

John of St. Thomas treats these as properties of the proposition. They hold between propositions with the same terms. Later logicians call them "immediate inferences."

The square of opposition

Pair	Relation	Rule
A and O; E and I	contradictories	never both true, never both false
A and E	contraries	never both true; both can be false
I and O	subcontraries	never both false; both can be true
A and I; E and O	subalterns	truth goes down; falsity goes up

So: if A is true, I is true; if I is false, A is false. A false universal leaves its particular "undetermined," and a true particular leaves its universal undetermined. Wherever the square forbids nothing, the answer is "undetermined." Subalternation is not properly opposition, but the square sets it out with the rest.

The contradictory

Change both quantity and quality: A to O, E to I, I to E, O to A.

Conversion

Exchange subject and predicate, keeping the quality and the truth.

- E and I convert **simply**: "No S are P" gives "No P are S"; "Some S are P" gives "Some P are S."
- A converts only **by limitation**, to I: "All S are P" gives "Some P are S," because the predicate of an affirmative is not distributed.

- O has no converse.

Obversion

Change the quality and replace the predicate with its complement, written "non-": "All S are P" gives "No S are non-P"; "No S are P" gives "All S are non-P"; "Some S are P" gives "Some S are not non-P"; "Some S are not P" gives "Some S are non-P." Use the contradictory term ("non-just"), not the contrary ("unjust"), since a man may be neither just nor unjust.

Contraposition

Obvert, convert, obvert. Both terms become complements and change places. "All S are P" gives "All non-P are non-S." "Some S are not P" gives "Some non-P are not non-S." E contraposes only by limitation. I has no contrapositive.

Equipollence: where the "not" goes

- "Not" **before** gives the contradictory: "Not every S is P" means "Some S are not P."
- "Not" **after the subject** of a universal gives the contrary: "Every S is not P" means "No S are P."
- "Not" **before and after** gives the subaltern: "Not every S is not P" means "Some S are P."

When the placement makes no equipollent, the answer is "none."

When nothing exists

Because affirmatives carry existence, obversion from the negative to the affirmative (E to A, O to I) and contraposition hold only when the terms stand for things that exist (Buridan). "No unicorn is a horse" is true, but "Every unicorn is a non-horse" is false. When the subject stands for nothing, the answer is "none."

The Modal Propositions

A modal proposition says not only that the predicate belongs to the subject but how it belongs. It has two parts: the **dictum**, what is said ("that Peter is running"), and the **mode**, said of the dictum ("it is possible").

The four modes

John of St. Thomas counts four: necessary, impossible, possible, and contingent. In this course the **contingent** is what can be and can also not be.

The four corners

Corner	Meaning	Like
necessary	cannot be false	"every"
impossible (necessary not)	cannot be true	"no"
possible	can be true	"some"
possible not	can be false	"some not"

Where the "not" goes

- A "not" **before the mode** moves to the contradictory corner: "not possible" is "impossible"; "not necessary" is only "possible not."
- A "not" **after the mode** denies only the dictum: "possible that he is not" is "possible not"; "necessary that he is not" is "impossible."

So "It is not possible that Peter is not running" comes to "It is necessary that Peter is running." Two expressions are **equipollent** when they come to the same corner. The commonest error is to read "not necessary" as "impossible."

The modal square

- Necessary and impossible are **contraries**: not both true; both false of what is contingent.

- Possible and possible not are **subcontraries**: not both false; both true of what is contingent.
- Necessary and possible not, and impossible and possible, are **contradictories**: always opposite.
- From the necessary to the possible the inference holds; from the possible to the necessary it does not. Truth goes down from the stronger mode; only falsity goes up.

Composite and divided sense

In the **composite sense** the mode governs the whole dictum taken together. "It is possible that the seated man walks" is false, since sitting and walking cannot be joined at once. In the **divided sense** the mode is said of the subject alone. "The seated man can walk" is true, since sitting does not take away the power to walk.

- "The one who is seated is necessarily seated": true composite, false divided.
- "A man is necessarily an animal": true in both, since it belongs by essence.
- "The blind man can see": true in neither, since blindness removes the power itself. So also "A man can be a stone."

Diagramming Syllogisms

A syllogism has three terms and three circles. The **minor term** is the subject of the conclusion; the **major term** is its predicate; the **middle term** appears in both premises and not in the conclusion.

How the diagram is built

- Each premise relates the middle term to one of the other two, so it is drawn on the two circles it names.
- Shade the universal premises first. Each one empties two cells, because the third circle cuts its region in two.
- Then put each particular premise's × in the one cell of its region that the shading leaves open.
- The conclusion is not drawn. Once the premises are drawn, the conclusion can be read from the diagram.

The marks

Shading: the region is empty. ×: the diagram's mark that something is here (I asserts it; for O it is only the diagram's mark). ⊗: something is here because an affirmative universal premise asserts that its subject exists.

When the ⊗ is needed

Sometimes the conclusion is particular though neither premise is, as in Darapti: "All M are P; all M are S; therefore some S are P." Shading alone does not show that anything exists. The existence comes from an affirmative premise, since an affirmative asserts that its subject exists. Draw the ⊗ in the only unshaded cell of the term whose existence the conclusion needs (in Darapti, M).

- If a particular premise already gives the ×, no ⊗ is drawn.
- A ⊗ never takes the place of a particular premise's ×.
- In Camestros and Camenos the term concerned appears only in a negative premise; the ⊗ is still drawn there.

The nine moods that take a ⊗

The five **weakened** moods, which conclude in particular what could be concluded universally: **Barbari, Celaront, Cesaro, Camestros, Camenos**. And four more: **Darapti, Felapton, Bramantip, Fesapo**. On the modern reading all nine are invalid. This course follows the traditional reading, on which all nine are valid.

Why

The premises are the reasons; the conclusion is already inside them. If we draw only the premises and the conclusion shows up in the picture, we can see that it follows.

The full list of valid moods and the general rules are given in the next primer (Exercise IX).

Syllogistic Form in the Abstract

A syllogism is judged by its form alone. The major premise comes first, then the minor, then the conclusion. Capital letters are general terms; lower-case letters are singular terms (one individual).

Terms and figure

The predicate of the conclusion is the **major term** (P); its subject is the **minor term** (S); the **middle term** (M) is in both premises and never in the conclusion. The figure is fixed by where the middle term stands:

Figure	Major premise	Minor premise
First	M P	S M
Second	P M	S M
Third	M P	M S
Fourth	P M	M S

A singular proposition such as "s is P" speaks of the whole of its subject, so it counts as universal, and its subject is distributed.

Distribution

A term is **distributed** when the proposition speaks of everything the term stands for.

- A distributes its subject only.
- E distributes both terms.
- I distributes neither.
- O distributes its predicate only.

Short form: the predicate of an affirmative is never distributed; the predicate of a negative always is.

The six general rules

1. The middle term must be distributed at least once. Otherwise: "undistributed middle."
2. A term distributed in the conclusion must be distributed in its premise. Otherwise: "illicit major" or "illicit minor."
3. From two negative premises nothing follows.
4. A negative premise requires a negative conclusion.
5. A negative conclusion requires a negative premise.
6. From two particular premises nothing follows.

A form that keeps all six is valid; a form that breaks any one is invalid. It also follows that a particular premise allows only a particular conclusion.

The 24 valid moods (traditional)

First: Barbara, Celarent, Darii, Ferio, Barbari, Celaront.
Second: Cesare, Camestres, Festino, Baroco, Cesaro, Camestros.
Third: Darapti, Disamis, Datisi, Felapton, Bocardo, Ferison.
Fourth: Bramantip, Camenes, Dimaris, Fesapo, Fresison, Camenos.

The vowels give the mood: Barbara is AAA in the first figure. You do not need to memorize the names; the rules are enough. A weakened conclusion ("Some S are P" where "All S are P" could be drawn) is also valid, because affirmatives carry existence.

Why the rules are so

The middle term joins the other two. If it is never taken in full, the two premises may be talking about different parts of it, and nothing is joined. A conclusion may not say more than its premises gave it.

EXERCISE X

Syllogistic Form in the Concrete

Here the syllogisms use ordinary English words, but we still judge the **form** only.

The key idea

- True premises and a true conclusion do **not** make an argument valid.
- False or silly premises do **not** make it invalid.
- Valid means: the conclusion must be true if the premises are, whatever they are about.

The surest method: replace each term with a letter and judge the form as in Exercise IX.

Reading the terms

- A proper name such as "Socrates" is a singular term. A singular proposition counts as universal, since it speaks of the whole of its subject.
- Plural or singular wording ("dogs," "a dog," "every dog") does not change the term.

The rules, in brief

Distribution: A distributes its subject; E both terms; I neither; O its predicate.

1. The middle term must be distributed at least once.
2. No term may be distributed in the conclusion unless it is distributed in its premise.
3. Nothing follows from two negative premises.
4. Nothing follows from two particular premises.
5. A negative premise requires a negative conclusion.
6. A negative conclusion requires a negative premise.

Example

"All men are mortal. Socrates is a man. So Socrates is mortal." Letters: all M are P; s is M; so s is P. The middle term "man" is distributed in the first premise. Valid.

"All dogs are animals. All cats are animals. So all cats are dogs." Every premise may be true, but the middle term "animals" is the predicate of two affirmatives and is never distributed. Invalid: undistributed middle.

Why

Logic studies the shape of reasoning. A good shape can carry false matter, and a bad shape can happen to land on a truth. We judge the shape.

EXERCISE XI

Drawing the Conclusion

Two premises are given. We find the conclusion that follows, in standard form, or answer "none."

Steps

1. **Find the middle term,** the one in both premises. The conclusion joins the other two terms and never contains the middle.
2. **Fix the quality.** If either premise is negative, the conclusion is negative. If both are affirmative, it is affirmative.
3. **Fix the quantity.** If either premise is particular, the conclusion is particular. Otherwise it may be universal, but a term may be distributed in the conclusion only if it was distributed in its premise. So sometimes only a particular conclusion follows.

Either remaining term may be the subject of the conclusion, as long as the rules are kept. Where a universal conclusion follows, the particular under it also follows and is accepted.

Standard form

"All S are P," "No S are P," "Some S are P," "Some S are not P."

Distribution reminder

A distributes its subject; E both; I neither; O its predicate.

When nothing follows

- Two negative premises.
- Two particular premises.
- The middle term is distributed in neither premise.
- Every possible conclusion would distribute a term its premise leaves undistributed.

Examples

"All men are mortal. All Greeks are men." The middle term is "men." Both affirmative and universal; "Greeks" is distributed in its premise. Conclusion: "All Greeks are mortal."

"All M are P. All M are S." Both universal, but S and P are each the predicate of an affirmative, so neither is distributed. Only a particular follows: "Some S are P."

"No M are P. Some M are not S." Two negatives: "none."

Why

The conclusion can be no stronger than the weaker premise, and it can say no more about a term than the premises said.

The Conjunctive and Disjunctive Syllogism

These syllogisms are built on whole propositions joined by a connective. The major premise is compound. The minor premise **posits** one member (asserts it) or **removes** it (denies it). The conclusion posits or removes the other member.

The conjunctive: "not both"

It denies that two propositions are true together: "No one can serve both God and Mammon."

- **Lawful:** posit one, remove the other. "Judas serves Mammon; therefore he does not serve God." This is the mood that denies by affirming (modus ponendo tollens), the only lawful mood of the conjunctive.
- **Fallacy:** remove one, posit the other. From "He does not serve Mammon" it does not follow that he serves God, because both members may be false together.

The disjunctive: "either ... or"

It says at least one member is true.

- **Lawful always:** remove one, posit the other. "The number is either odd or even; it is not odd; therefore it is even." This is the mood that affirms by denying (modus tollendo ponens).
- **Posit one, remove the other:** lawful only when the disjunction is **strict**.

Strict and broad

- **Strict**, marked "but not both": exactly one member is true, so both moods are lawful.
- **Broad**, marked "or perhaps both": at least one is true. From "He is either clever or lucky, or perhaps both" and "He is clever," nothing follows about his luck.

Always read the marker before judging.

Members that are whole propositions

To remove a member that is a categorical proposition is to assert its contradictory: A and O, E and I. So "Some B are not C" removes "All B are C," and "No D are F" removes "Some D are F." Whatever the members are, the mood alone decides.

Why

"Not both" only forbids two being true together; it says nothing about both being false. "Either ... or" only promises at least one; it says nothing against both unless it adds "but not both."

The Hypothetical Syllogism

A conditional asserts a connection, not a fact: "If it has rained, the ground is wet." The part after "if" is the **antecedent**; the part it supports is the **consequent**. A conditional is true when the consequent really follows from the antecedent, even if neither is true in fact.

The two lawful moods

- **The mood that affirms by affirming** (modus ponendo ponens): posit the antecedent, and the consequent follows. "It has rained; therefore the ground is wet."
- **The mood that denies by denying** (modus tollendo tollens): remove the consequent, and the antecedent is removed. "The ground is not wet; therefore it has not rained." This holds because the antecedent cannot be true without the consequent.

The two fallacies

- **Affirming the consequent:** "The ground is wet; so it rained." It fails because the consequent may be true on other grounds, as a burst pipe would also wet the ground.
- **Denying the antecedent:** "It did not rain; so the ground is not wet." It fails for the same reason.

A conditional asserts the connection in one direction only, from antecedent to consequent. It does not assert the reverse, even when the reverse happens to be true.

Negative parts

In "If Socrates is not a B, then Socrates is a C," to posit the antecedent is to assert "Socrates is not a B," and to remove it is to assert "Socrates is a B."

When a part is a whole categorical proposition, to remove it is to assert its contradictory: "Some B are not C" removes "All B are C." In every case the mood decides.

Summary

Second premise	Result
Posits the antecedent	Valid: posit the consequent
Removes the consequent	Valid: remove the antecedent
Posits the consequent	Fallacy
Removes the antecedent	Fallacy

Why

"If A, then B" means A never comes without B. But B might come with other things too. So finding B does not find A, and losing A does not lose B.

The Mixed Consequence

This exercise mixes the conditional, conjunctive, and disjunctive. We type what follows, or "none."

The lawful moods

Premise	Lawful	Concludes nothing
"If A, then B"	posit A, so B (ponendo ponens); remove B, so not A (tollendo tollens)	posit B; remove A
"Not both A and B"	posit one, so not the other (ponendo tollens)	remove one
"Either A or B"	remove one, so the other (tollendo ponens)	posit one, when broad
"Either A or B, but not both"	remove one, so the other; posit one, so not the other	

A broad disjunction is marked "or perhaps both"; positing one member of it removes nothing.

Writing the answer

- When the premises are about one subject, write the answer about that subject: "Socrates is a C" or "Socrates is not a C."
- Removing a negative part asserts the positive: if the premise begins "If Socrates is not a B," then removing that antecedent gives "Socrates is a B."
- When a part is a whole categorical proposition, denying it asserts its contradictory: A and O, E and I. If the conclusion removes "All B are C," write "Some B are not C."
- Where the mood is fallacious, the answer is "none."

Why

Each connective promises one thing only. "If" promises the consequent whenever the antecedent is there. "Not both" forbids two together. "Either ... or" promises at least one. Use only what is promised.

Arguments in Ordinary Speech

Arguments of every kind are stated in ordinary English. We find the syllogism and judge it.

1. Find the conclusion

"Therefore," "so," "hence," "it follows that" mark the conclusion. "Since," "because," "for," "after all" mark a premise. The conclusion often comes first: "No oaks are shrubs, since every oak is a tree and no tree is a shrub."

2. Put each premise in standard form

- **A:** "Every S is a P," "Whatever is an S is a P," "If something is an S, then it is a P," "S are always P."
- **E:** "S are never P," "Not a single S is a P," "There is no S that is a P."
- **I:** "There are S that are P," "Certain S are P," "A few S are P."
- **O:** "Not all S are P," "Not every S is a P." Also "Few S are P," which chiefly denies.
- "All S are not P" may mean "No S are P" or only "Some S are not P" ("All that glitters is not gold" means the second). Read it twice.
- "Only P are S" means "All S are P." So do "none but" and "no one except."
- With no sign of quantity, the sense decides: "Dogs are mammals" means all; "Dogs were barking in the alley" means some.
- "All dogs bark" becomes "All dogs are things that bark."
- A singular subject ("Socrates") counts as universal.

3. Judge it

Categorical: the middle term distributed at least once; no term distributed in the conclusion unless in its premise; nothing from two negatives or two particulars; a negative premise needs a negative conclusion and the reverse.

Compound: "If ... then" concludes by positing the antecedent or removing the consequent; affirming the consequent is the common error. "Not both" concludes only by positing one to remove the other. "Either ... or" concludes by removing one to posit the other, and by positing one to remove the other only when strict ("but not both").

Modal: the conclusion may not claim a stronger mode than follows. Same mode in both premises gives that mode. In the first figure, a necessary major with a plain minor gives a necessary conclusion. A contingent premise (first figure: the major; third figure: either) gives only a contingent conclusion. In the second figure two contingent premises give nothing.

The Modal Syllogism

A modal syllogism has premises that say how the predicate belongs. Three modes are used here.

- **Necessary:** it cannot be otherwise.
- **Assertoric** (plain): it simply is so.
- **Contingent:** it can be, and can also not be.

The answer is: necessarily, assertorically, contingently, or nothing follows.

First test: the form

If the syllogism breaks a general rule, nothing follows in any mode. In quality and quantity the conclusion follows the weaker premise: a negative premise needs a negative conclusion, a particular premise a particular one.

The mode (after Aristotle)

- Both premises necessary: necessary conclusion.
- Both plain: plain conclusion.
- First figure, necessary major with plain minor: **necessary**. "Every man is necessarily an animal; every runner is a man; therefore every runner is necessarily an animal."
- Plain major with necessary minor: only **plain**, because the major says only that the predicate belongs, not that it must.

Contingent premises

- First figure: a contingent major with a contingent or plain minor gives a **contingent** conclusion.
- Third figure: two contingent premises, or one contingent and one plain, give a **contingent** conclusion.
- Second figure: two contingent premises give **nothing**, whatever their quantity or quality.

The exercise asks only about these cases.

Why the necessary major wins

What belongs necessarily to everything under the middle term belongs necessarily to whatever falls under it. The minor only places the runner under "man"; the major says what "man" must be.

Theophrastus held that the conclusion always follows the weaker premise in mode too. The course follows Aristotle, with the medieval commentators and St. Thomas.

The Chain of Syllogisms

A chain of syllogisms (a **polysyllogism**) links several syllogisms so that the conclusion of one is a premise of the next.

> All B are C; all A are B; therefore all A are C.
> All C are D; therefore all A are D.

In each link after the first, one new premise is stated, and the conclusion just drawn is the other premise. When the in-between conclusions are left unstated, the chain is called a **sorites**; the chains here state every conclusion.

How to judge it

- Judge link by link.
- In each link, the middle term is the term its two premises share and its conclusion leaves out.
- Apply the general rules to each link.

The general rules

1. The middle term must be distributed at least once.
2. No term may be distributed in the conclusion unless it is distributed in its premise.
3. Nothing follows from two negative premises.
4. Nothing follows from two particular premises.
5. A negative premise requires a negative conclusion, and a negative conclusion requires a negative premise.

Distribution: A distributes its subject; E both; I neither; O its predicate.

The chain rule

The whole chain is valid only if every link is valid. One bad link makes the whole chain invalid, even when the last conclusion looks right.

Why

A chain is only as strong as its weakest link. If one conclusion was never proved, everything built on it stands on nothing.

The Matter and the Form

Every argument can be judged two ways. Its **form** is the arrangement of its terms. Its **matter** is what the premises say.

The three answers

- **Valid** means it is impossible for the premises to be true and the conclusion false, whatever the terms stand for.
- **Sound** means valid with true premises. A sound argument proves its conclusion.
- **Valid & Sound:** good form, true premises.
- **Valid, Unsound:** good form, but a false premise. It proves nothing. This is the commonest imitation of knowledge.
- **Invalid:** bad form. It fails even when every premise and the conclusion are true.

Judge the form first, by the general rules; then test each premise.

Formal and material logic

Formal logic studies the arrangement of the terms and whether the conclusion follows. Material logic studies whether the premises are true and how each science reasons about its own subject.

Sound is not yet demonstration

A demonstration is a valid syllogism whose premises are true, better known than the conclusion, and give the necessary causes of the conclusion, so it shows why the conclusion must be. This exercise asks only about validity and truth.

The four matters and surprising truths

Premises come from everyday things, Euclid's mathematics, the virtues (Aristotle and St. Thomas), and the soul and living things (De

Anima). Key terms are given above each argument. Test a surprising premise against the definitions, not first impressions:

- For Euclid, an isosceles triangle has two sides alone equal, so no equilateral triangle is isosceles.
- The unit is not a number. A point is not a magnitude.
- For St. Thomas, a passion is not a habit; the theological virtues observe no mean; prudence is intellectual in what it is, but counted with the moral virtues because it concerns conduct.
- In the De Anima, the oyster is an animal without sight.

Why

To prove something you need both a good road (form) and a true starting point (matter). Either alone is not enough.

The Enthymeme

An enthymeme is a syllogism with a premise (or sometimes the conclusion) left unspoken, which the hearer is trusted to supply. "Socrates is mortal, for he is a man" leaves out "All men are mortal."

Vocabulary

- **First-order:** the major premise is left unsaid.
- **Second-order:** the minor premise is left unsaid.
- A premise is left unsaid for brevity, because it is obvious, or because it is weak and would not be granted. So the hidden premise is where an argument hides its weakness. Supply it and examine it.
- Restoring a valid form does not make the argument sound; the supplied premise may be false.

Supplying the tacit premise

1. Find the two terms of the conclusion: its predicate is the major term, its subject the minor term.
2. Find the middle term: the term of the stated premise that is not in the conclusion.
3. Join the middle term to the term of the conclusion that the stated premise leaves out, in a quantity and quality that make a valid syllogism. Any premise that does so is accepted.

"All A are C, since all B are C": the tacit premise is "All A are B."

Sometimes nothing can complete it, because the stated premise is particular or negative and the conclusion claims more than any added premise could support: "All A are C, for some B are C." Answer "none."

The right direction

For "Socrates is mortal, for he is a man," the premise needed is "All men are mortal," not "All mortals are men" (which leaves the middle

undistributed). "Some men are mortal" is too weak to conclude anything about Socrates.

Signs

Many examples argue from signs.

- A **necessary sign** concludes of necessity: "She has given birth, for she has milk," with "Whoever has milk has given birth."
- A **refutable sign** does not: "It must have rained, for the streets are wet" needs "Whenever the streets are wet, it has rained," which is false, since a street-sweeper also wets the streets. The true premise "Whenever it rains, the streets are wet" only gives affirming the consequent.

Dialectic

Dialectic reasons from what is **probable**: what seems so to everyone, or to most, or to the wise (and among the wise, to all or to the most eminent). It reaches a **reasoned opinion**, well supported but still open to challenge. It stands between rhetoric and demonstration: more than persuasion, less than proof.

The standing of a premise

- **Demonstrable:** can be proved; a matter of science, not dialectic.
- **Granted by everyone:** probable in the fullest sense; may simply be assumed. A self-evident principle ("Every whole is greater than any one of its parts") is granted by everyone; it is certain but not proved.
- **Granted by the learned:** still probable and usable, but expect to defend it.
- **One person's opinion:** perhaps true, but not yet something an opponent can be asked to grant.

Places and maxims

Logic has two arts: **judging** an argument (the Analytics) and **finding** one (the Topics). A **place** (locus, topic) is a standing relation, such as genus to species, from which an argument is drawn. Its **maxim** is the rule the argument depends on.

Place	Maxim
Definition	What fits the definition fits the thing, and what does not, does not.
Wider kind (genus)	What belongs to the whole family belongs to each kind in it; what the family lacks, each kind lacks.
Narrower kind (species)	What the narrower kind is said of, the wider is said of too.
Property	Where the property is, the thing is; where it is absent, the thing is absent.

Opposites	What holds of one opposite, the contrary holds of the other.
More and less	If it holds where less expected, it holds all the more where more expected.
Likeness	What holds in one case holds in a like case, so far as they are alike.
Whole and parts	What holds of the whole holds of the parts, when it holds of the whole by reason of the parts.
Cause and end	Where the cause is, the effect follows; a thing is judged by what it is for.
Same root (conjugates)	If justice is good, acting justly is good.
The name	What the name means tells something of the thing, though not always, and never alone.
Authority	What those who know agree on may be granted, until better reason appears. The weakest place: it moves us by who says it, not by what makes it so.

Answering an objection

The one who puts the case is the opponent (opponens); the one who answers is the respondent (respondens). Four answers:

1. **Distinguish the term** (the distinguo): a word is true in one sense and false in another; grant one, deny the other.
2. **Deny the premise:** it is simply false.
3. **Grant it all and deny the consequence:** the matter is sound, but the form is not.
4. **Grant the whole objection** and state the thesis again more narrowly.

Dialectic argues toward the better view; a testing disputation only finds out whether someone knows.

The Fallacies

A fallacy is an argument that looks sound and is not. Following De fallaciis (attributed to St. Thomas), each fallacy has two causes: the **cause of the appearance** (what makes it look good) and the **cause of the failure** (what breaks it). The answer removes the cause of failure. Every argument depends on a rule, usually unspoken; a fallacy depends on a rule that is false but looks true.

Six in the words

Why six: ambiguity is threefold (actual, potential, apparent), and two of these divide into word and phrase.

- **Equivocation** (actual, word): one word, two meanings. "Dog": the animal and the star.
- **Amphiboly** (actual, phrase): one phrase, two readings. "The book of Aristotle": written or owned?
- **Accent** (potential, word): a change of stress changes the claim.
- **Composition** (potential, phrase): takes words together that are true only apart. "A seated man can walk."
- **Division** (potential, phrase): takes words apart that are true only together. "Two and three are five."
- **Figure of speech** (apparent): words of the same form treated as signifying the same way.

Seven outside the words

The language is fine, but two things that agree in some way are taken as simply one.

- **Accident:** takes a passing feature for the thing. "I know the man who is coming; Coriscus is the man coming; so I know Coriscus."
- **In a certain respect, taken flatly:** what is true in some respect taken as true without limit (modern: hasty generalization).
- **Missing the point** (ignorance of the refutation): proves something else. A real refutation contradicts the very thing said,

in the same respect and at the same time. Straw man and red herring are species of it.

- **Begging the question:** assumes what was to be proved.
- **Treating what is not the cause as the cause:** blames a premise that did nothing toward an impossible conclusion.
- **The consequent:** reads a one-way link as two-way. "The ground is wet, so it rained." Modern: affirming the consequent, and the undistributed middle.
- **Many questions asked as one:** hides a second question, so either answer grants it (modern: loaded question).

Disputations and the sophist

Four disputations: **demonstrative** (teacher and learner, certain knowledge), **dialectical** (reasoned opinion from the probable), **testing** (does he really know?), **sophistical** (to seem wise and win; "sophistical" means apparent wisdom).

The sophist's five aims, each offending a different discipline: make you **contradict yourself** (metaphysics), grant something **plainly false** (natural science and mathematics), grant something **absurd** (dialectic), say something **ungrammatical** (grammar), or **babble** (rhetoric).